

Lecture (That was given as Homework)

Charging and Discharging a Capacitor

Charging a Capacitor

Whenever we connect an uncharged capacitor with a voltage source, it is exponentially charged until the voltage developed across the capacitor becomes equal and opposite to the voltage of the source.

Let us connect one capacitor of capacitance C in series with a resistor of resistance R . We also connect this series combination of capacitor and resistor with a battery of voltage V through a push switch S (Figure 1).

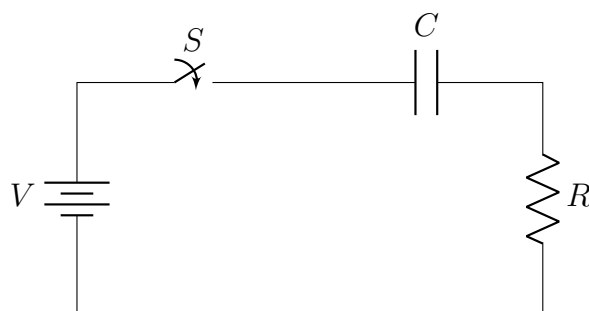


Figure 1: Circuit Diagram of a simple series RC circuit.

When we push the switch, the capacitor is uncharged, hence no voltage developed across the capacitor. Thus, the capacitor will behave as a short circuit. The current through the circuit will only be limited by resistance R . That initial current is V/R . Now slowly the voltage is being developed across the capacitor, and this developed voltage is in opposite of the polarity of battery. As a result, the current in the circuit gradually decreased. It will be gradually decreased to zero, when the voltage across the capacitor becomes equal and opposite of the voltage of battery.

The voltage is gradually increased across the capacitor during charging. Let us consider the rate of increase of voltage across the capacitor is dv/dt at any instant t . The current through the capacitor at that instant is:

$$i = dQ/dt \quad \text{and Charge, } Q = CV$$

$$i = C \frac{dv}{dt}$$

Applying Kirchhoff's Voltage Law, in the circuit at that instant, we can write:

$$V = v + Ri$$

$$V = v + RC \frac{dv}{dt} \quad \left[\because i = C \frac{dv}{dt} \right]$$

Rearranging the terms for integration:

$$\Rightarrow V - v = RC \frac{dv}{dt}$$

$$\Rightarrow \frac{dt}{CR} = \frac{dv}{V - v}$$

Integrating both sides:

$$\int \frac{dt}{CR} = \int \frac{dv}{V - v}$$

$$\frac{t}{CR} = -\ln(V - v) + A$$

Now, at the time of switching on the circuit, voltage across the capacitor was zero. That means, $v = 0$ at $t = 0$. Putting these values in the above equation, we get:

$$\frac{0}{CR} = -\ln(V - 0) + A$$

$$\Rightarrow 0 = -\ln V + A$$

$$\Rightarrow A = \ln V$$

After getting the value of A, we can rewrite the above equation as:

$$\frac{t}{CR} = -\ln(V - v) + \ln V$$

$$\Rightarrow \frac{t}{CR} = \ln \left(\frac{V}{V - v} \right)$$

Exponentiating both sides:

$$\Rightarrow \frac{V}{V - v} = e^{\frac{t}{CR}}$$

$$\Rightarrow V - v = V e^{-\frac{t}{CR}}$$

$$\Rightarrow v = V \left(1 - e^{-\frac{t}{CR}} \right) \quad \text{--- (1)}$$

This is the expression of voltage across the capacitor during charging.

Now, we know that, the current is:

$$i = C \frac{dv}{dt} = C \frac{d}{dt} \left[V \left(1 - e^{-\frac{t}{CR}} \right) \right]$$

$$= CV \frac{d}{dt} \left(1 - e^{-\frac{t}{CR}} \right) = CV \left(0 - e^{-\frac{t}{CR}} \cdot \left(-\frac{1}{CR} \right) \right)$$

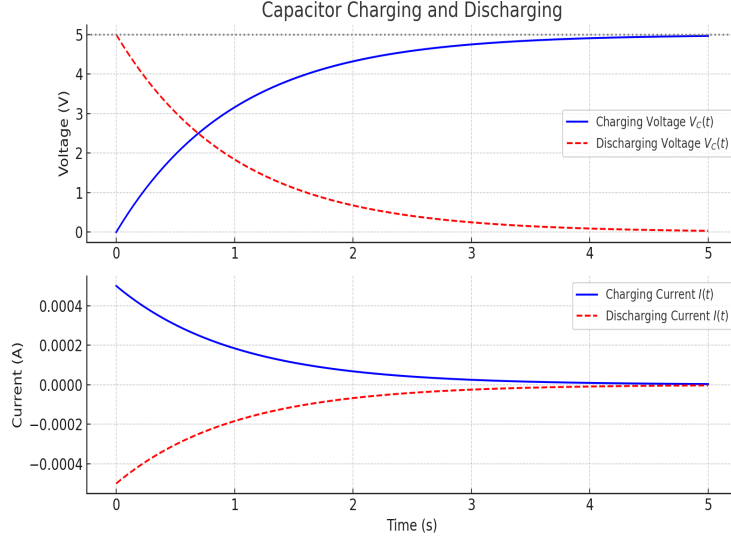
$$= CV \left(\frac{1}{CR} e^{-\frac{t}{CR}} \right) = \frac{V}{R} e^{-\frac{t}{CR}} \quad \text{--- (2)}$$

This is the expression of charging current i , during the process of charging.

The current and voltage of the capacitor during charging is shown in Figure 2.

Capacitor Charging and Discharging

This diagram illustrates the charging and discharging behavior of a capacitor through a resistor.



- The top plot shows the voltage across the capacitor: it rises exponentially during charging and decays exponentially during discharging.
- The bottom plot shows the current: it decreases over time in both charging and discharging, but in opposite directions.

Figure 2: Top graph shows voltage across capacitor over time. Bottom graph shows the discharging current over time.

Transient Behavior of Capacitor

At the beginning, this current will be maximum and after certain time the current will become zero. The duration in which current changes in capacitor is known as transient period. The phenomenon of charging current or other electrical quantities like voltage, in capacitor is known as transient.

To understand transient behavior of capacitor, let us draw an RC circuit as shown below: (Refers to a circuit diagram showing a voltage source V , a switch S , a resistor R , and a capacitor C in series, with a current loop $i(t)$ and voltage across capacitor $V_c(t)$ indicated.)

Now, if the switch S is suddenly closed, the current starts flowing through the circuit. Let the current at any instant be $i(t)$. Also consider the voltage developed at the capacitor at that instant is $v_c(t)$. Hence, by applying Kirchhoff's Voltage Law, in that circuit we get, from equation (1):

$$v = V \left(1 - e^{-\frac{t}{CR}} \right)$$

Now if we put $t = CR$ (which is the time constant) in the above equation, we get:

$$v_c = V(1 - e^{-CR/CR}) = V(1 - e^{-1}) = 0.632V$$

This CR (or product of resistance and capacitance of RC series circuit) is known as the **time constant** of the circuit. So, time constant of an RC circuit, is the time for which voltage developed or dropped across the capacitor is 63.2% of the supply voltage. This definition of time constant only holds good when the capacitor was initially uncharged.

Now at any instant, current through the circuit will be, from equation (2):

$$i(t) = \frac{V - v_c(t)}{R} = \frac{V - V[1 - e^{-t/(CR)}]}{R} = \frac{V}{R}e^{-t/(CR)} = I_o e^{-t/(CR)}$$

Now when $t = CR$, the circuit current:

$$i = I_o e^{-1} = 0.367I_o$$

So at the instant when, current through the capacitor is 36.7% of the initial current, is also known as time constant of the RC circuit. The time constant is normally denoted with τ . Hence, $\tau = CR$.

Discharging a Capacitor

Please refer to the circuit diagram (Fig.1) showing a charged capacitor C, a resistor R, and a switch S in series, forming a closed loop for discharge. Let us connect a charged capacitor of capacitance C farad in series with a resistor of resistance R ohms. We then short circuit this series combination through switching on the push switch as shown.

As soon as the capacitor is short circuited, it starts discharging. Let us assume, the voltage of the capacitor at fully charged condition is V volt. As soon as the capacitor is short circuited, the discharging current of the circuit would be V/R ampere. But after the instant switching on, that is after $t = 0$, the current through the circuit is:

$$i = C \frac{dv}{dt}$$

As per Kirchhoff's Voltage Law, we get:

$$\begin{aligned} v + iR &= 0 \\ \Rightarrow v + RC \frac{dv}{dt} &= 0 \\ \Rightarrow v &= -RC \frac{dv}{dt} \\ \Rightarrow \frac{dv}{v} &= -\frac{dt}{RC} \end{aligned}$$

Integrating both sides, we get:

$$\begin{aligned} \int \frac{dv}{v} &= - \int \frac{dt}{RC} \\ \ln v &= -\frac{t}{RC} + A \end{aligned}$$

Where, A is the constant of integral and, at $t = 0$, $v = V$.

$$\ln V = -\frac{0}{RC} + A$$

$$\Rightarrow A = \ln V$$

After calculating value of A, we get:

$$\ln v = -\frac{t}{RC} + \ln V$$

$$\Rightarrow \ln \left(\frac{v}{V} \right) = -\frac{t}{RC}$$

Exponentiating both sides:

$$\Rightarrow \frac{v}{V} = e^{-\frac{t}{RC}}$$

$$\Rightarrow v = V e^{-\frac{t}{RC}} \quad \text{--- (3)}$$

This is the expression of voltage across the capacitor during discharging.

We know from KVL of the circuit, $i = -\frac{v}{R}$. Substituting the expression for v :

$$i = -\frac{1}{R} V e^{-\frac{t}{RC}}$$

$$= -\frac{V}{R} e^{-\frac{t}{RC}} = -I_o e^{-\frac{t}{RC}} \quad \left[\text{Here, } I_o = \frac{V}{R} \right] \quad \text{--- (4)}$$

If we plot these discharging current and voltage in graph, we get: (Please refer to Fig 2, where the graph shows discharging current and voltage decreasing exponentially over time, with the current being negative.)

Transient During Discharging a Capacitor

Now, suppose the capacitor is fully charged. i.e. voltage at capacitor is equal to the voltage of source. Now if the voltage source is disconnected and instead two terminals of the battery are short circuited, the capacitor will start discharging. This means, unequal distribution of electrons between two plates will be equalized through the short circuit path. The process of equalizing electrons concentration in two plates will continue until the voltage at capacitor becomes zero. This process is known as discharging of capacitor. Now we will examine the transient behavior of capacitor during discharging.

In the circuit diagram shown in Fig.1 with a resistor R and a capacitor C in series, imagine a with a current loop $i(t)$ and voltage across capacitor $\vartheta_c(t)$ representing the discharge circuit.

Now, from the above circuit by applying Kirchhoff Current Law (likely meant KVL based on context), we get from equation (3), by applying $t = \tau = RC$, we get:

$$\vartheta_c(\tau) = V e^{-1} = 0.368V$$

Again, circuit current at that time i.e. $\tau = RC$:

$$|i(\tau)| = \frac{\vartheta_c(\tau)}{R} = \frac{0.368V}{R} = 0.368I_o$$

Thus at time constant of capacitor, both capacitor voltage, ϑ_c and current i are reduced to 36.8% of their initial value.